

Lie Theory: Chapter 1 Homework

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1. Let $\mathbb{R}_\vee^3$ be the Lie algebra defined in Section 1.2. Prove that if $x \cdot y$ denotes the dot product of vectors $x, y \in \mathbb{R}^3$, then

$$x \wedge (y \wedge z) = (x \cdot z)y - (z \cdot y)x \quad \forall x, y \in \mathbb{R}^3.$$

2. Check that

$$[e_{ij}, e_{kl}] = \delta_{jk}e_{il} - \delta_{il}e_{kj},$$

where δ is the Kronecker delta, defined by $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ otherwise.

3. Prove that the set e_{ij} for $i \neq j$ together with the set $e_{ii} - e_{i+1, i+1}$ for $1 \leq i < n$ forms a basis for $\mathfrak{sl}(n, F)$.

4. Prove that an ideal of a Lie algebra is always a subalgebra.

5. Consider the adjoint homomorphism map defined (on page 4) for all $x, y \in L$ by

$$(\text{ad } x)(y) := [x, y].$$

(a) Prove that ad is linear.

(b) Prove that ad is a homomorphism.

(c) Prove that the kernel of ad is the center of L .

6. Prove that $\text{Der}(A)$ is a vector subspace of $\mathfrak{gl}(A)$.

7. Let L_1 and L_2 be two abelian Lie algebras. Show that L_1 and L_2 are isomorphic if and only if they have the same dimension.

8. Find the structure constants of $\mathfrak{sl}(2, F)$ with respect to the basis given by the matrices

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

9. Prove that $\mathfrak{sl}(2, \mathbb{C})$ has no non-trivial ideals.

10. Let S be an $n \times n$ matrix with entries in a field F . Define

$$\mathfrak{gl}_S(n, F) = \{x \in \mathfrak{gl}(n, F) : x^T S = -Sx\}.$$

(a) Show that $\mathfrak{gl}_S(n, F)$ is a Lie subalgebra of $\mathfrak{gl}(n, F)$.

(b) Find $\mathfrak{gl}_S(2, \mathbb{R})$ if $S = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.

11. Let A be an algebra and let $\delta : A \rightarrow A$ be a derivation. Prove that δ satisfies the Leibniz Rule

$$\delta^n(xy) = \sum_{r=0}^n \binom{n}{r} \delta^r(x) \delta^{n-r}(y) \quad \forall x, y \in A.$$